

Cybersecurity Program Schedule

20-Day Intensive Training Calendar (July 27 – August 24, 2026)

Program:	5th Batch CTC Cybersecurity Program	Total Duration:	4 Weeks (20 Interaction Days)
Dates:	July 27, 2026 – August 24, 2026	Daily Schedule:	09:00 - 17:00 (Mon – Fri)
Track:	Cybersecurity Specialized Curriculum	Capstone:	2-Day Offensive/Defensive CTF (Days 19–20)

20-DAY CYBERSECURITY SCHEDULE

Day	Date	Cybersecurity Module & Technical Topics
WEEK 1: FOUNDATIONS (DAYS 1–5)		
Day 1	Jul 27	Kickoff, Ethics & Linux Fundamentals Ethics sign-off, VM installation, host-only network isolation, Linux CLI, file permissions, SSH.
Day 2	Jul 28	Networking Fundamentals & Bash Scripting OSI & TCP/IP, IP/subnetting, DNS, ports/services, Wireshark traffic inspection, Nmap, Bash automation.
Day 3	Jul 29	Cryptography & Web Fundamentals Hashing (SHA-256), TLS/HTTPS, PKI, Base64, HTTP request/response inspection via Burp Suite.
Day 4	Jul 30	Programming for Security (Python & C) Python automation scripts, C memory management, pointers, compilation, and basic binary analysis.
Day 5	Jul 31	Computer Architecture & Assembly Intro CPU registers, stack/heap, x86/x64 assembly instruction set, PE/ELF executable structures.
WEEK 2: PENETRATION TESTING & OFFENSIVE SECURITY (DAYS 6–10)		
Day 6	Aug 03	Reconnaissance, Scanning & Enumeration Pentest methodologies, scoping, Nmap deep scanning, SMB & SNMP enumeration, target profiling.
Day 7	Aug 04	Web Pentesting I: Injection & Input Flaws OWASP flaws, path traversal, file upload bypass, SQL Injection (SQLi), OS Command Injection.
Day 8	Aug 05	Web Pentesting II: Auth, Access & Logic IDOR, business logic vulnerabilities, SSRF, XSS, CSRF, JWT security, multi-stage attack chaining.
Day 9	Aug 06	Privilege Escalation & Active Directory Intro Linux SUID/cron priv-esc, Windows token impersonation, AD architecture, Kerberos, LDAP, NTLM.
Day 10	Aug 07	Active Directory Attacks & Pivoting BloodHound attack paths, Kerberoasting, AS-REP roasting, mimikatz, lateral movement, Ligolo-ng pivoting.
WEEK 3: REVERSE ENGINEERING & MALWARE ANALYSIS (DAYS 11–15)		
Day 11	Aug 10	Reverse Engineering: Static Analysis Disassembly fundamentals, Ghidra/IDA Free decompilation, PE headers, cross-references, solving crackmes.
Day 12	Aug 11	Reverse Engineering: Dynamic Analysis & Debugging x64dbg debugging, memory breakpoints, anti-debug/anti-VM bypasses, UPX unpacking, binary patching.
Day 13	Aug 12	Malware Analysis: Basic Static & Dynamic Malware taxonomy, triaging with PEstudio/Procmon/Regshot/Process Hacker, INetSim network simulation.
Day 14	Aug 13	Malware Techniques & Benign Loader Dev

Day	Date	Cybersecurity Module & Technical Topics
		Process injection, persistence mechanisms, API hashing, building a benign C loader, artifact detection.
Day 15	Aug 14	Advanced Malware, Forensics & YARA Volatility 3 memory forensics, C2 config extraction, YARA rule writing, MITRE ATT&CK mapping.
WEEK 4: DEFENSE, MOBILE & CAPSTONE (DAYS 16–20)		
Day 16	Aug 17	Blue Team & SOC Fundamentals Defense-in-depth, SIEM concepts, Wazuh deployment, agent enrollment, custom alert rule creation.
Day 17	Aug 18	DFIR & Log Analysis Incident response lifecycle, Sysinternals, Windows winevtx & Linux log analysis, timeline reconstruction.
Day 18	Aug 19	Mobile Security Fundamentals Android architecture, JADX static analysis, Frida/objectection dynamic instrumentation, client bypasses.
Day 19	Aug 20	Capstone CTF - Day 1 (Offensive Phase) Team-based offensive challenges: web exploitation, Active Directory compromise, reverse engineering.
Day 20	Aug 21	Capstone CTF - Day 2 (Defense) & Closing Blue-team log triage, incident report writing, final team presentations, awards, and certification.
Event	Aug 24	Program Showcase & Graduation Exhibition showcase; capstone project demonstrations, awards, and certificate distribution.

DAILY INTERACTION STRUCTURE

Time Block	Focus Area	Activities & Objectives
09:00 - 10:30	Morning Lecture I	Concept lecture + live technical demonstration
10:30 - 10:45	Morning Break	Refreshment break
10:45 - 12:15	Morning Lecture II	Concept continuation + deep-dive tool walkthrough
12:15 - 13:15	Lunch Break	Lunch
13:15 - 15:15	Afternoon Lab I	Guided hands-on lab exercise in isolated VM environment
15:15 - 15:30	Afternoon Break	Short break
15:30 - 16:45	Afternoon Lab II	Independent lab execution / challenge solving
16:45 - 17:00	Wrap-up & Checkpoint	Daily checkpoint submission (flag, script, or write-up) + recap

Daily Deliverable Requirement: Every interaction day terminates with a concrete deliverable (e.g., captured flag, automation script, or brief technical write-up) to verify mastery before advancing to subsequent topics.